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[22EC52]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

FUNDAMENTALS OF IOT

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	5 x2=10		
1.	a.	Mention any two applications of IoT in Agriculture.	L1	CO1	02
	b.	Explain in brief how M2M differ from traditional human-to-machine communication?	L2	CO2	02
	c.	Name two key components of the Raspberry Pi that enable it to function as a mini-computer.	L1	CO3	02
	d.	Name two core services offered by AWS for IoT.	L1	CO4	02
	e.	What is Thermistor?	L1	CO5	02
		PART-B	6 X 5=30		
2.	a.	Explain the characteristic features of IoT.	L2	CO1	05
		OR			
	b.	Mention different methods of link layer with corresponding standards.	L2	CO1	05
3.	a.	Illustrate the key differences between TCP and UDP.	L2	CO1	05
		OR			
	b.	Explain how does IoT fit into M2M communication?	L2	CO2	05
4.	a.	Discuss the roles of M2M area networks and communication networks in the end-to-end architecture of M2M systems?	L2	CO2	05
		OR			
	b.	Explain Simple Network Management Protocol with a suitable diagram.	L2	CO2	05
5.	a.	Explain the various connectivity modules used for designing IoT model with their uses.	L2	CO3	05
		OR			
	b.	Differentiate between I2C and SPI communication protocols.	L3	CO3	05
6.	a.	Illustrate the steps to get started with Linux on a Raspberry Pi. Also, discuss on preparation, installation, and initial setup procedures.	L2	CO3	05
		OR			
	b.	Explain the salient characteristic features of cloud computing.	L2	CO4	05
7.	a.	Discuss the working principle of stepper motor with a neat diagram.	L2	CO5	05
		OR			

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	b.	With a neat diagram explain the significance of transistor in controlling the high-power devices using Raspberry Pi.	L2	CO5	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Describe the various components required for design of IoT model and briefly discuss on Level-1 IoT model.	L2	CO1	**10**
		OR			
	b.	Explain any three IoT Communication Modules with a suitable diagram.	L2	CO1	**10**
9.	**a.**	Show the structure of four-layer IoT Protocol and explain the function of each layer.	L2	CO1	**10**
		OR			
	b.	With a neat diagram, illustrate the architecture of NETCONF protocol.	L2	CO2	**10**
10.	**a.**	Discuss the role of SDN in IoT emphasizing on management, scalability, and security.	L2	CO2	**10**
		OR			
	b.	Describe the key features of YANG (Yet Another Next Generation), and discuss about its relationship with NETCONF.	L2	CO2	**10**
11.	**a.**	Briefly explain the key features, main components, and operating system of Raspberry Pi.	L2	CO3	**10**
		OR			
	b.	With a neat circuit and python code, illustrate the interfacing of an LED and Button with Raspberry Pi.	L2	CO3	**10**
12.	**a.**	What are IoT communication APIs, and explain how do they contribute to the functionality of IoT systems?	L2	CO4	**10**
		OR			
	b.	Mention the key functions of web servers for IoT and explain how do they facilitate communication and data management between IoT devices and users?	L2	CO4	**10**
13.	**a.**	Illustrate the interfacing of PIR motion sensor with a Raspberry Pi with a suitable code and circuit diagram.	L2	CO6	**10**
		OR			
	b.	What steps are involved in using a relay with a Raspberry Pi to manage the AC device? Describe the role of the relay and include a simple Python script to facilitate the control.	L2	CO6	**10**